

Gaussians in the City: Enhancing 3D Scene Reconstruction under distractors with Text-guided Segmentation and Inpainting

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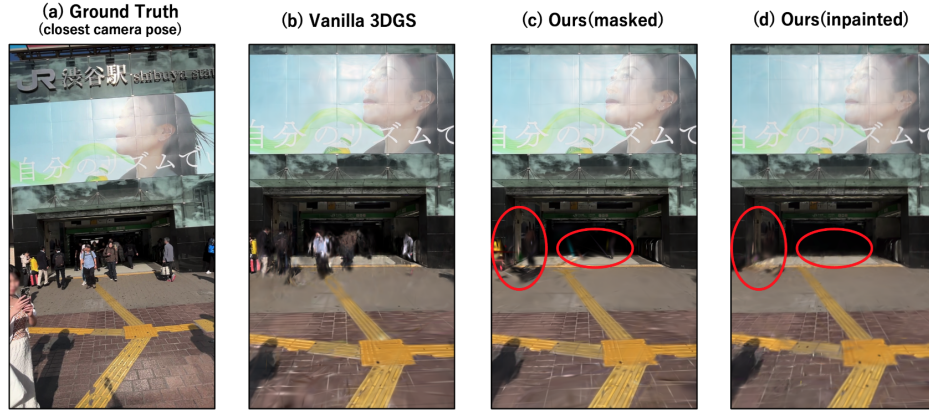


Figure 1: Comparison of Rendered Images: (a) ground truth image in dataset (b) rendered image from vanilla 3DGS from a new camera pose (c) ours after initial optimization (d) ours after inpainting and second optimization. Red circles illustrate defects and artifacts around heavily masked regions are suppressed in our method with inpainting.

Abstract

We propose a novel method to reconstruct 3D Gaussian scene from images that contain both static and dynamic distractors, e.g., vehicles and pedestrians, by using text-guided segmentation and inpainting techniques. Our approach generates masks for distractors, detects heavily masked regions and inpaints them to suppress defects and artifacts for scene optimization while ignoring distractors.

CCS Concepts

• Computing methodologies → Image-based rendering.

Keywords

3D Reconstruction, Text-Guided Distractor Removal, 3D Inpainting

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1 INTRODUCTION

3D Gaussian Splatting (3DGS)[Kerbl et al. 2023] has made significant advancements in novel view synthesis in accuracy and speed; however, costly data collection remains a barrier to practical applications. Many existing studies assume scenes without static or dynamic distractors, such as vehicles and pedestrians. Traffic control is required for large-scale urban or tourist sites captures.

Several studies have addressed this by (1) separately learning static and dynamic scenes to remove dynamic objects[Zhang et al. 2024], and (2) applying masks using networks to detect distractor pixels trained on time-series uncertainty[Ren et al. 2024]. However, these methods struggle with stationary distractors, such as standing person, and can be unstable in areas like floors that are often obscured. Additionally, users cannot specify distractors directly.

In this paper, we propose a robust 3DGS method that operates effectively even in highly crowded environments. Our method allows users to identify both dynamic and static obstacles through text while preventing artifacts in excessively masked, data-deficient areas by using a pre-trained inpainting model with a detection algorithm for heavily masked regions, as shown in Figure1.

2 METHODS

When using inpainting for novel view synthesis, ensuring 3D consistency is essential. Weber et al.[Weber et al. 2024] reported that arranging multiple viewpoint images on a grid results in more consistent 3D completion than inputting each image individually into 2D inpainting diffusion models. Also, including an unmasked image

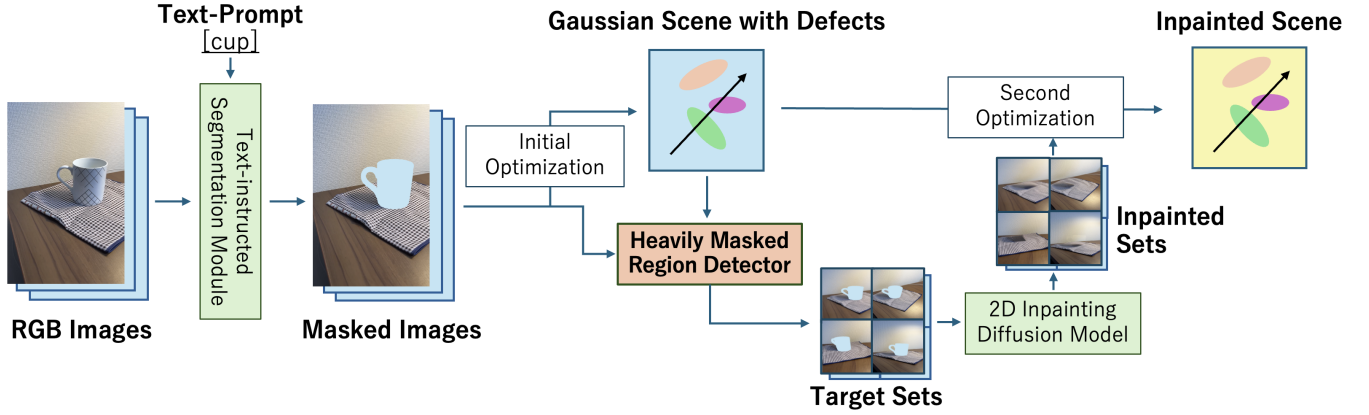


Figure 2: System Overview: Our system generates distractor masks from text prompts and optimizes 3D Gaussian scene with a loss function that ignores masked pixels. Then, our detection algorithm identifies heavily masked regions and generates sets of four images for a 2D inpainting model, avoiding inpainting overuse. Finally, optimization resumes on the updated dataset.

in the set guides the inpainting process. Inspired by their work, we propose an algorithm applying 3D-consistent inpainting only for heavily masked regions, as excessive use compromises consistency.

Figure 2 illustrates an overview of our system. Starting with images and camera poses, we first generate masks for the distractors specified by text prompts using the lang-segment-anything module [Medeiros 2023], which combines an object detection and a transformer-based segmentation models. Then, we perform the initial Gaussian optimization with modified loss function ignoring masked pixels. At this stage, artifacts and defects are observed around heavily masked regions. Algorithm 1 summarizes our inpainting procedure. In order to identify heavily masked regions, we calculate projected positions in pixel space for each Gaussian and camera pose. Then, we randomly select a Gaussian projected onto masked regions on four or more camera poses and onto unmasked regions by six or fewer camera poses. Four images are selected from camera poses rendering the Gaussian, including one projecting the Gaussian onto unmasked regions for guiding the inpainting, if available. These images are then arranged on a grid and inpainted by using LaMa [Suvorov et al. 2021] inpainting model. This process is repeated to update the dataset, achieving 3D consistency with minimal inpainting. We restart the optimization on the updated images, initializing the scene with the center points of the Gaussians.

Algorithm 1 Proposed Inpainting Algorithm

Require: I, M, C : Images, Masks and Camera Parameters

Require: G : Gaussians

Require: k : number of iterations

```

1: for  $j = 1$  to  $k$  do
2:   for each  $g \in G$  do
3:      $C_{on}(g), C_{off}(g) \leftarrow \text{find\_C\_render\_on/off\_mask}(g, C, M)$ 
4:   end for
5:   Randomly select  $g_t$  from  $\text{len}(C_{off}) \leq 6$  AND  $\text{len}(C_{on}) \geq 4$ 
6:    $C_{inpaint} \leftarrow C_{off}(g) + (C_{on}(g) \text{ until } \text{len}(C_{inpaint}) = 4)$ 
7:    $I, M \leftarrow \text{inpaint\_selected\_images}(I, M, C_{inpaint})$ 
8: end for

```

3 RESULTS & FUTURE WORK

We collected 153 images in a busy area, estimated camera parameters using COLMAP [Schönberger and Frahm 2016], and optimized Gaussian scenes by vanilla 3DGS and our method. Positive prompts ["human", "person", "belongings"] and negative prompts ["billboard", "signboard"] were used for mask generation. Figure 1 shows (a) an image in the dataset, (b) rendered images from vanilla 3DGS (*vanilla*), ours after (c) initial and (d) second optimizations (*masked*, *inpainted*). (b) *Vanilla* contains artifacts from distractors. While (c) *masked* contains defects and artifacts around heavily masked regions highlighted with red circles, (d) *inpainted* successfully suppressed them. Future work will include more detailed analysis and an interactive data collection system to mitigate heavily masked regions.

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